

LAB 6 STATIC ROUTING

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TASK 1: ROUTER IP CONFIGURATION

R1 IP: 10.0.0.1 - eth0/0

R1 IP: 20.0.0.1 - eth0/1

The diagram shows a network topology with a central router R1. R1 has two interfaces: eth0/0 with IP 10.0.0.1 and eth0/1 with IP 20.0.0.1. A laptop labeled 'Win-1' is connected to R1's eth0/0 interface. The CLI window for R1 shows the following configuration:

```
R1(config-if)#ip address 10.0.0.1 255.0.0.0
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#int eth0/1
R1(config-if)#ip address 20.0.0.1 255.0.0.0
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#exit
R1#
*Jul 16 04:16:49.723: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/8 is directly connected, Ethernet0/0
L    10.0.0.1/32 is directly connected, Ethernet0/0
C    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/1
L    20.0.0.1/32 is directly connected, Ethernet0/1
R1#
```

R2 IP: 20.0.0.2 - eth0/0

R2 IP: 30.0.0.1 - eth0/1

The diagram shows a network topology with two routers, R2 and R3. R2 has two interfaces: eth0/0 with IP 20.0.0.2 and eth0/1 with IP 30.0.0.1. R3 has two interfaces: eth0/1 with IP 30.0.0.2 and eth0/0 with IP 40.0.0.1. A laptop labeled 'Win-2' is connected to R3's eth0/0 interface. The CLI window for R2 shows the following configuration:

```
R2(config)#int eth0/0
R2(config-if)#ip address 20.0.0.2 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#int eth0/1
R2(config-if)#ip address 30.0.0.1 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#exit
R2#
*Jul 16 04:28:19.822: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/0
L    20.0.0.2/32 is directly connected, Ethernet0/0
C    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.1/32 is directly connected, Ethernet0/1
R2#
```

R3 IP: 30.0.0.2 – eth0/1

R3 IP: 40.0.0.1 – eth0/0

The screenshot shows a network simulator interface. On the left, a terminal window for R3 displays the following configuration commands and output:

```
R3 Wizard
R3(config)#int eth0/1
R3(config-if)#ip address 30.0.0.2 255.0.0.0
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#int eth0/0
R3(config-if)#ip address 40.0.0.1 255.0.0.0
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#exit
R3#
*Jul 16 04:33:35.938: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR
Gateway of last resort is not set

  30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
  C    30.0.0.0/8 is directly connected, Ethernet0/1
  L    30.0.0.2/32 is directly connected, Ethernet0/1
  C    40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
  C    40.0.0.0/8 is directly connected, Ethernet0/0
  L    40.0.0.1/32 is directly connected, Ethernet0/0
R3#
```

On the right, a network diagram shows three routers (R2, R3, R4) and a host (Win-2). R2 is connected to R3 via their Ethernet0/1 interfaces. R3 is connected to R4 via their Ethernet0/0 interfaces. R4 is connected to Win-2 via its Ethernet0/0 interface. The IP addresses are labeled on the connections: 30.0.0.1 on R2's Ethernet0/1, 30.0.0.2 on R3's Ethernet0/1, 40.0.0.1 on R3's Ethernet0/0, and 40.0.0.10 on Win-2's Ethernet0/0. The interface on R4 is labeled E0.

TASK 2: CREATING STATIC ROUTES IN THE ROUTER

STATIC ROUTE R1: 30.0.0.0/8 20.0.0.2 STATIC ROUTE R1: 40.0.0.0/8 20.0.0.2

The screenshot shows a network simulator interface. On the left, a network diagram shows a host (Win-1) connected to a router (R1) via their Ethernet0/0 interfaces. The IP address 10.0.0.10 is labeled on Win-1's Ethernet0/0 interface. The interface on R1 is labeled E0/0. The IP address 10.0.0.1 is labeled on R1's Ethernet0/0 interface. The interface on R1 is labeled E0/1. The IP address 20.0.0.1 is labeled on R1's Ethernet0/1 interface.

On the right, a terminal window for R1 displays the following configuration commands and output:

```
R1(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2
R1(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2
R1(config)#ip route 40.0.0.0 255.0.0.0 20.0.0.2
R1(config)#exit
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR
Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
  C    10.0.0.0/8 is directly connected, Ethernet0/0
  L    10.0.0.1/32 is directly connected, Ethernet0/0
  C    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
  C    20.0.0.0/8 is directly connected, Ethernet0/1
  L    20.0.0.1/32 is directly connected, Ethernet0/1
  S    30.0.0.0/8 [1/0] via 20.0.0.2
  S    40.0.0.0/8 [1/0] via 20.0.0.2
R1#
```

STATIC ROUTE R2: 10.0.0.0/8 20.0.0.1 STATIC ROUTE R2: 40.0.0.0/8 30.0.0.2

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
R2(config)#ip route 40.0.0.0 255.0.0.0 30.0.0.1
^Invalid next hop address (it's this router)
R2(config)#ip route 40.0.0.0 255.0.0.0 30.0.0.2
R2(config)#exit
R2#
*Jul 16 04:30:48.435: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
       a - application route
       * - replicated route, % - next hop override, p - overrides from PBR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
S    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/0
L    20.0.0.2/32 is directly connected, Ethernet0/0
C    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.1/32 is directly connected, Ethernet0/1
S    40.0.0.0/8 [1/0] via 30.0.0.2
R2#
```

Static Route

Physical

01 : 00 : 00 START

STATIC ROUTE R3: 10.0.0.0/8 30.0.0.1 STATIC ROUTE R3: 20.0.0.0/8 30.0.0.1

```
R3#
C    40.0.0.0/8 is directly connected, Ethernet0/0
L    40.0.0.1/32 is directly connected, Ethernet0/0
R3#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip route 10.0.0.0 255.0.0.0 30.0.0.1
R3(config)#ip route 20.0.0.0 255.0.0.0 30.0.0.1
R3(config)#exit
R3#
*Jul 16 04:36:07.256: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
       a - application route
       * - replicated route, % - next hop override, p - overrides from PBR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 30.0.0.1
S    20.0.0.0/8 [1/0] via 30.0.0.1
C    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.2/32 is directly connected, Ethernet0/1
C    40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    40.0.0.0/8 is directly connected, Ethernet0/0
L    40.0.0.1/32 is directly connected, Ethernet0/0
R3#
```

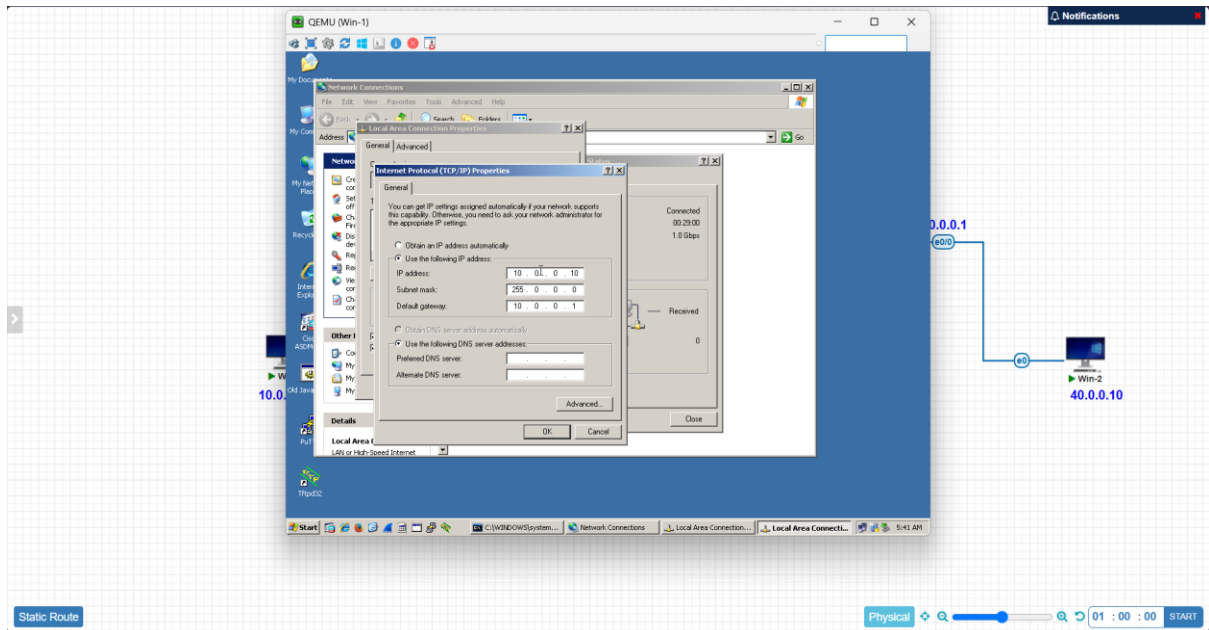
Static Route

Physical

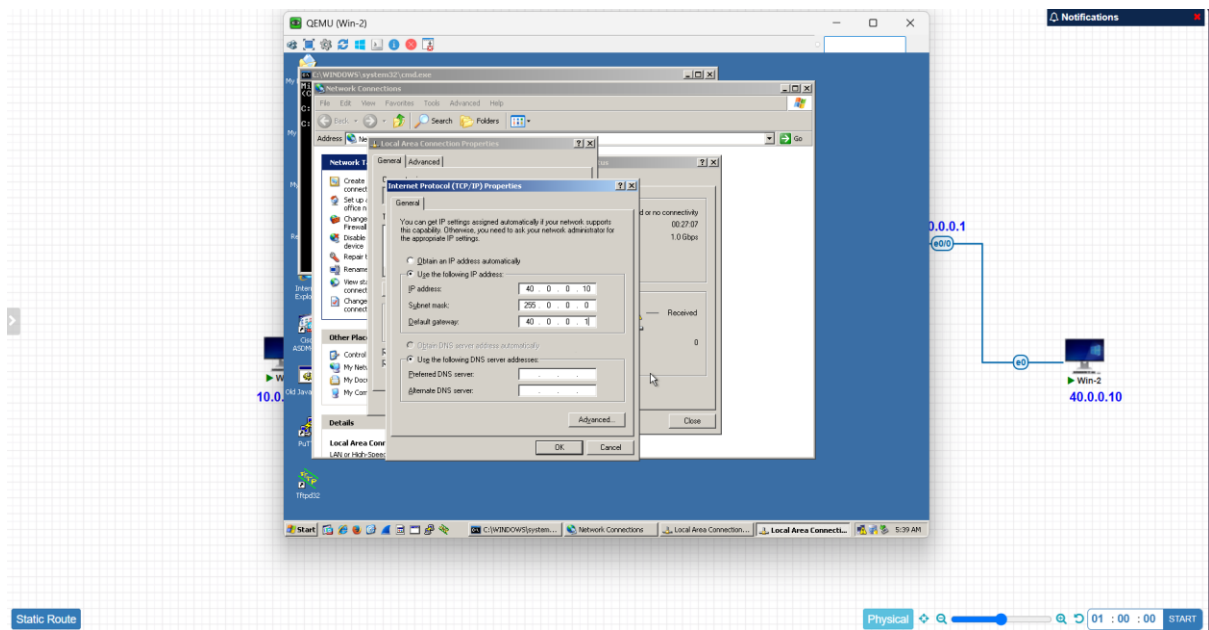
01 : 00 : 00 START

TASK 3: ASSIGNING IP TO PCs

PC 1: 10.0.0.10

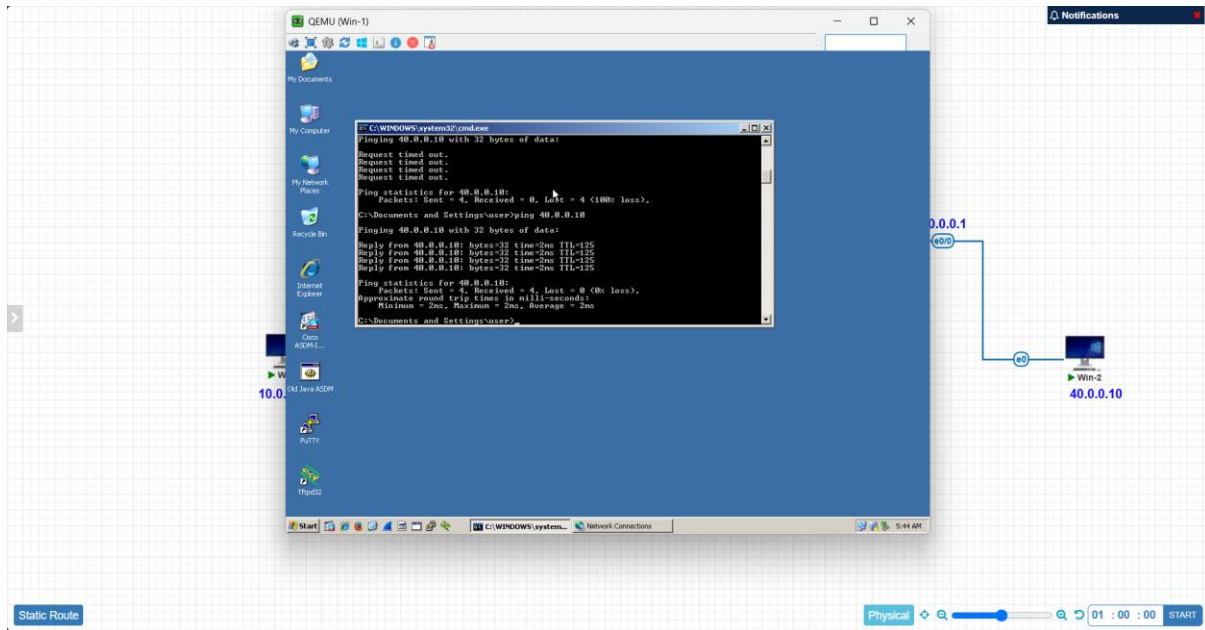


PC 2: 40.0.0.10



TASK 4: VERIFYING DATA FLOW USING PING

FROM PC1 TO PC2



FROM PC2 TO PC1

